

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems. (L4, L5)

Assessment Tool: Viva Voce

Weightage: 10%

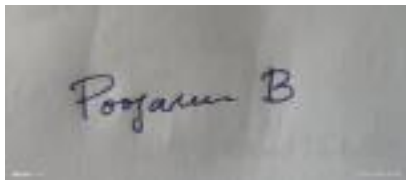
QUESTIONS:

Total Marks: 20

Code of Conduct:

I POOJASREE B (192411091) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

A photograph of a handwritten signature in blue ink on a light-colored background. The signature appears to be 'Poojasree B'.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively

1. Define SOAP.

2. Define WSDL.
3. What is an API?
4. Define XML Schema.
5. What is a Web Service?
6. Define REST.
7. What is a Protocol?
8. Define Endpoint in web services.
9. What is a Service in web applications?
10. What is the difference between SOAP and REST?
11. What is the role of WSDL in web services?
12. What is JSON and how is it used in APIs?
13. What is HTTP and why is it used in web services?
14. What is meant by request and response in web communication?
15. What is the importance of XML in web services?
16. What is the purpose of an API endpoint?
17. What is the difference between HTTP and HTTPS?
18. What is meant by RESTful services?
19. What is a namespace in XML?
20. What is the role of a client and server in web services?

1. SOAP

SOAP stands for **Simple Object Access Protocol**. It is a protocol used for communication between different applications over a network. SOAP messages are mainly written in **XML**, which provides a structured format for exchanging information. It is commonly used in enterprise applications where security and reliability are important.

2. WSDL

WSDL stands for **Web Services Description Language**. It is an XML-based language used to describe a web service. It specifies the operations provided by the service, the input and output messages, data types, and the location of the service. It helps clients understand how to communicate with the web service.

3. API

API stands for **Application Programming Interface**. It is a set of rules and methods that allows one software application to communicate with another. APIs allow applications to request data or services without knowing how the internal system works. For example, a weather application can use a weather API to obtain weather information.

4. XML Schema

An **XML Schema (XSD)** is used to define the structure and rules of an XML document. It specifies the elements, attributes, data types, and relationships that can be present in the document. It also helps validate whether an XML document follows the required structure.

5. Web Service

A **web service** is a software component that enables different applications to communicate and exchange data over a network. It allows applications developed using different programming languages or platforms to interact with each other. SOAP and REST are commonly used to implement web services.

6. REST

REST stands for **Representational State Transfer**. It is an architectural style used to design web services and APIs. REST uses standard HTTP methods such as **GET, POST, PUT, and DELETE** to perform operations on resources. REST APIs commonly use JSON because it is lightweight and easy to process.

7. Protocol

A **protocol** is a set of rules that defines how data is transmitted and received between two systems. It specifies how communication should take place so that both systems understand each other. Examples include **HTTP, HTTPS, FTP, and TCP/IP**.

8. Endpoint

An **endpoint** is a specific URL or address through which a web service or API can be accessed. Each endpoint usually represents a particular resource or operation. For example, <https://example.com/api/students> can be an endpoint used to access student information.

9. Service

A **service** is a specific functionality provided by a software application that can be accessed by users or other applications. It performs a particular task and provides a result. Examples include **login service, payment service, booking service, and order service**.

10. SOAP and REST

SOAP and REST are both used for communication between applications, but they work differently. **SOAP is a protocol** that mainly uses XML and follows a strict messaging structure. **REST is an architectural style** that commonly uses JSON and HTTP methods. SOAP is more complex and suitable for structured enterprise applications, while REST is simpler, lightweight, and widely used for web and mobile APIs.

11. Role of WSDL

WSDL provides detailed information about a web service. It describes the **operations available, input and output parameters, data types, communication method, and service endpoint**. It acts like a technical description or guide that allows a client application to understand and use the web service correctly.

12. JSON

JSON stands for **JavaScript Object Notation**. It is a lightweight and human-readable format used to store and exchange data. JSON represents information using objects and key-value pairs. It is widely used in REST APIs to send data from clients to servers and receive data from servers.

Example:

```
{
  "name": "Pooja",
  "age": 20
}
```

13. HTTP

HTTP stands for **HyperText Transfer Protocol**. It is a communication protocol used to transfer data between a client and a server over the web. Web services use HTTP to send requests and responses. Common HTTP methods include **GET for retrieving data, POST for creating data, PUT for updating data, and DELETE for removing data**.

14. Request and Response

A **request** is a message sent by a client to a server asking for information or requesting an operation. The **response** is the message returned by the server after processing the request. For example, a client may send a GET request for student details, and the server responds with the student's information.

15. Importance of XML

XML provides a **structured, readable, and platform-independent** method for representing and exchanging data. It allows different applications to understand and process the same data. XML is especially important in **SOAP web services**, where SOAP messages are formatted using XML.

16. API Endpoint

An API endpoint provides a specific access point to a particular resource or operation in an API. It is usually represented by a URL. For example, `/api/students` may return a list of students, while `/api/students/10` may return information about student 10. Therefore, endpoints help clients access specific API functionalities.

17. HTTP and HTTPS

HTTP transfers information between a client and server without encrypting the communication. **HTTPS (HTTP Secure)** uses **TLS encryption** to protect the data being transferred. HTTPS provides better security and protects sensitive information such as passwords, payment details, and personal data.

18. RESTful Services

RESTful services are web services that follow the principles of REST. They represent data as **resources**, with each resource identified using a URL. HTTP methods are used to perform operations on these resources. RESTful services are generally **simple, scalable, flexible, and lightweight**, making them popular for web and mobile applications.

19. XML Namespace

An **XML namespace** is used to uniquely identify elements and attributes in an XML document. It prevents naming conflicts when elements with the same name come from different sources. Namespaces are usually identified using a unique URI and are especially useful when combining multiple XML vocabularies.

20. Client and Server

The **client** is the application that sends a request for data or a service. The **server** receives the request, processes it, performs the required operation, and sends a response back to the client. For example, in an online shopping application, the mobile or web application acts as the client, while the backend system acts as the server.